



IoT-Based Smart Shuttle Management System for University Campus.



UNIVERSITY SHUTTLE: A DAILY ESSENTIAL

1) Quick and Convenient Way



1) Quick, reliable transit to campus. Destination reached!

2) Save Transportation Costs



2) Save on expensive private transit.

3) Extra Time with Friends



3) Extra social time and connection on the ride.

4) Arrive on Time During Exams



4) Avoid delays during high-stakes deadlines.

- For university students, shuttle service is an essential part of daily life.
- It provides a quick and convenient way to reach their destination.
- It also helps students save transportation costs.
- Using the shuttle gives students extra time to spend with their friends.
- During exams and project deadlines, it helps students arrive on time and avoid unnecessary delays.

Introduction

Motivation

Long Waits

This motivated us to develop a live vehicle tracking system.

Crowded & Confused

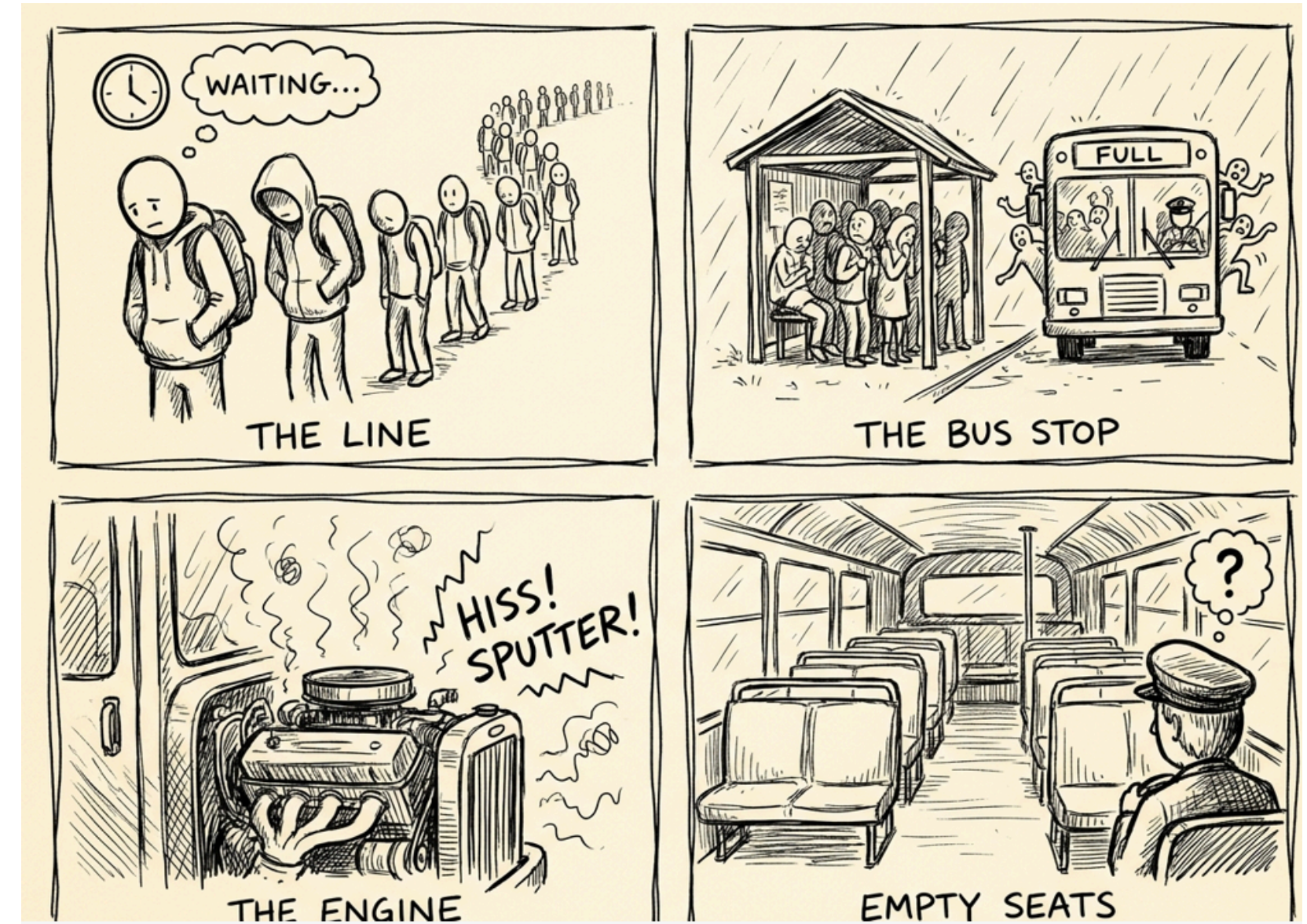
Motivated us to develop a system that will automatically count students

Bus malfunctions & Environmental impact

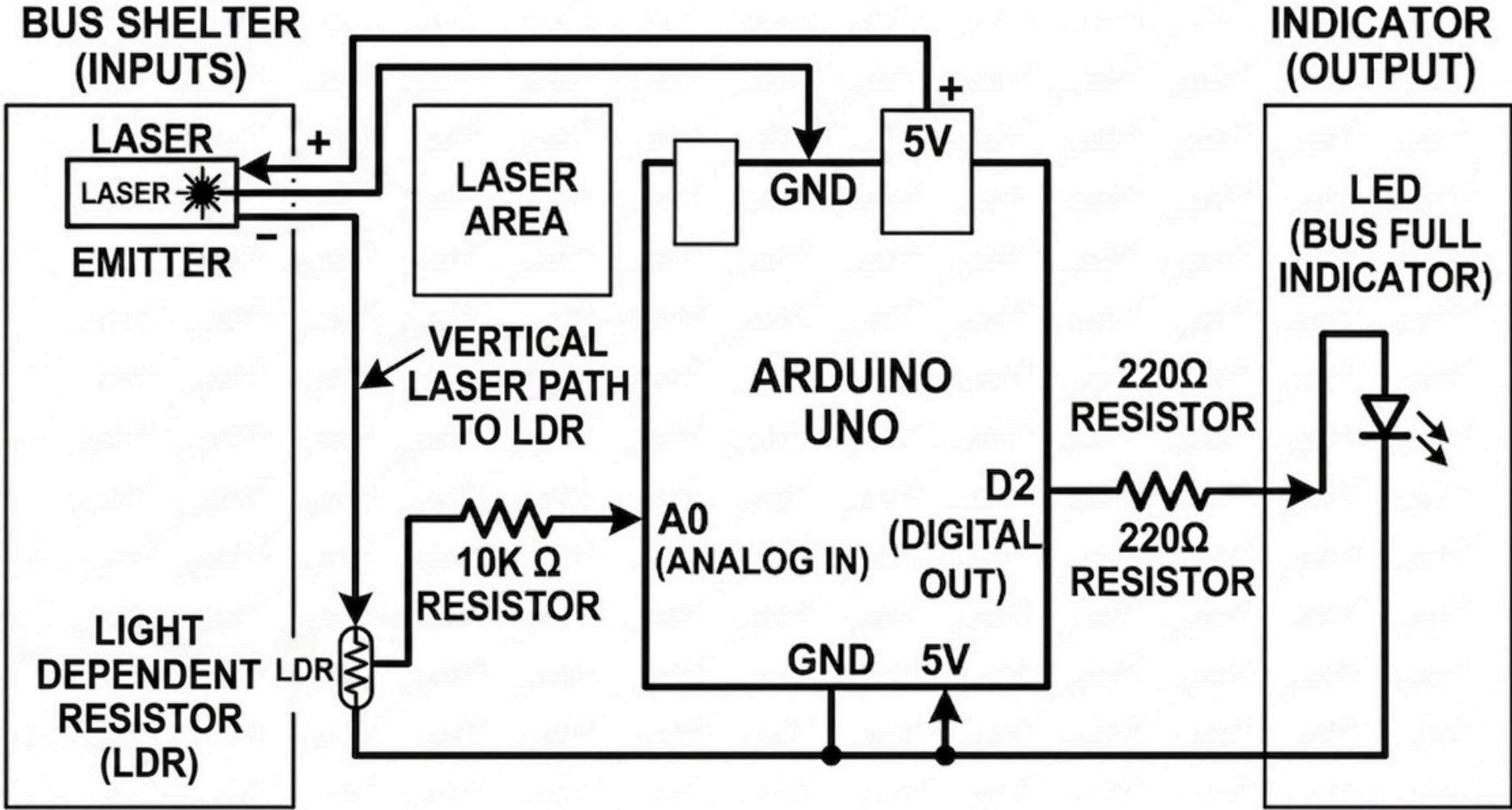
Motivated us to develop a smoke and environmental monitoring system.

Manual Checking

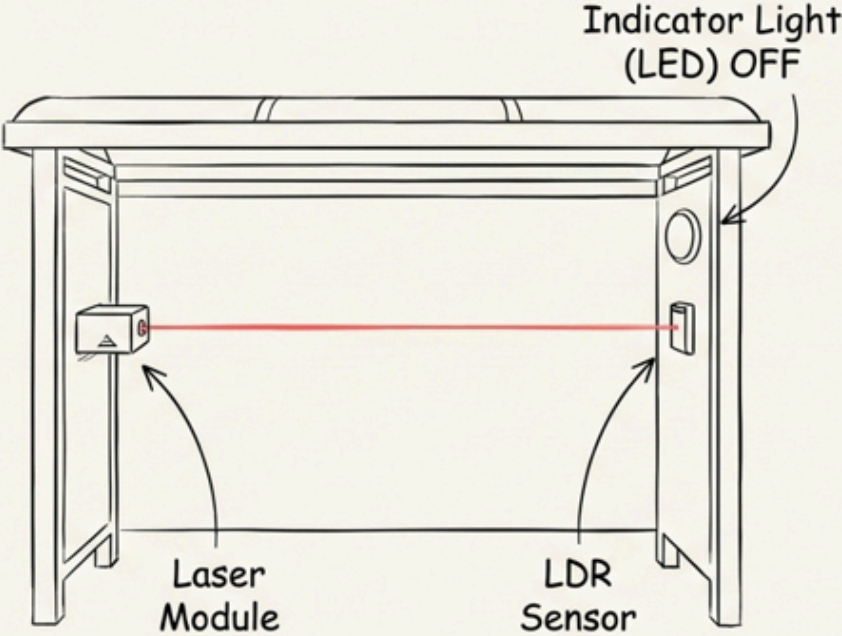
Motivated us to build a system that updates the driver about the crowd and bus status.



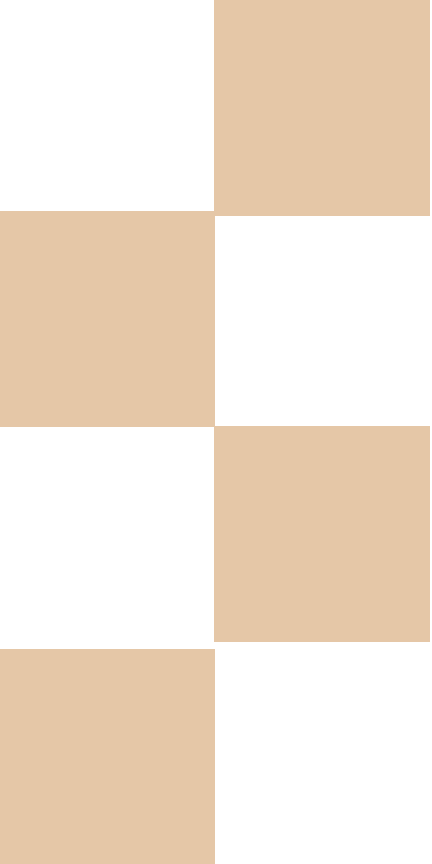
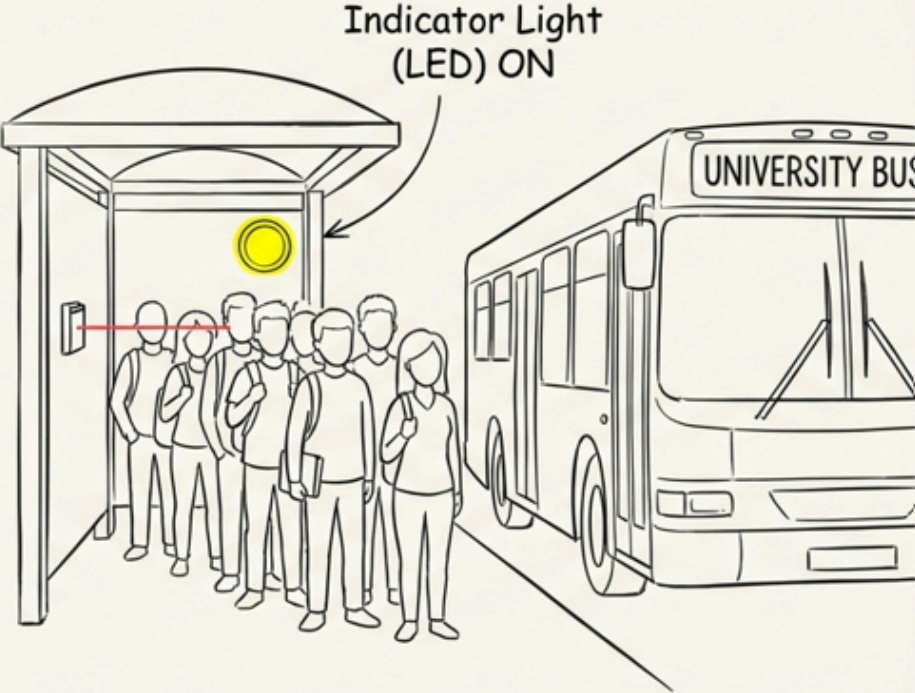
Feature 1



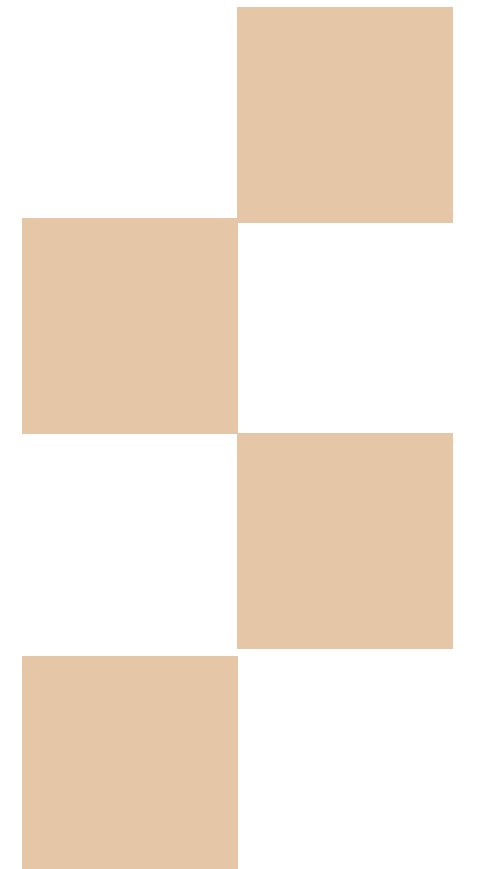
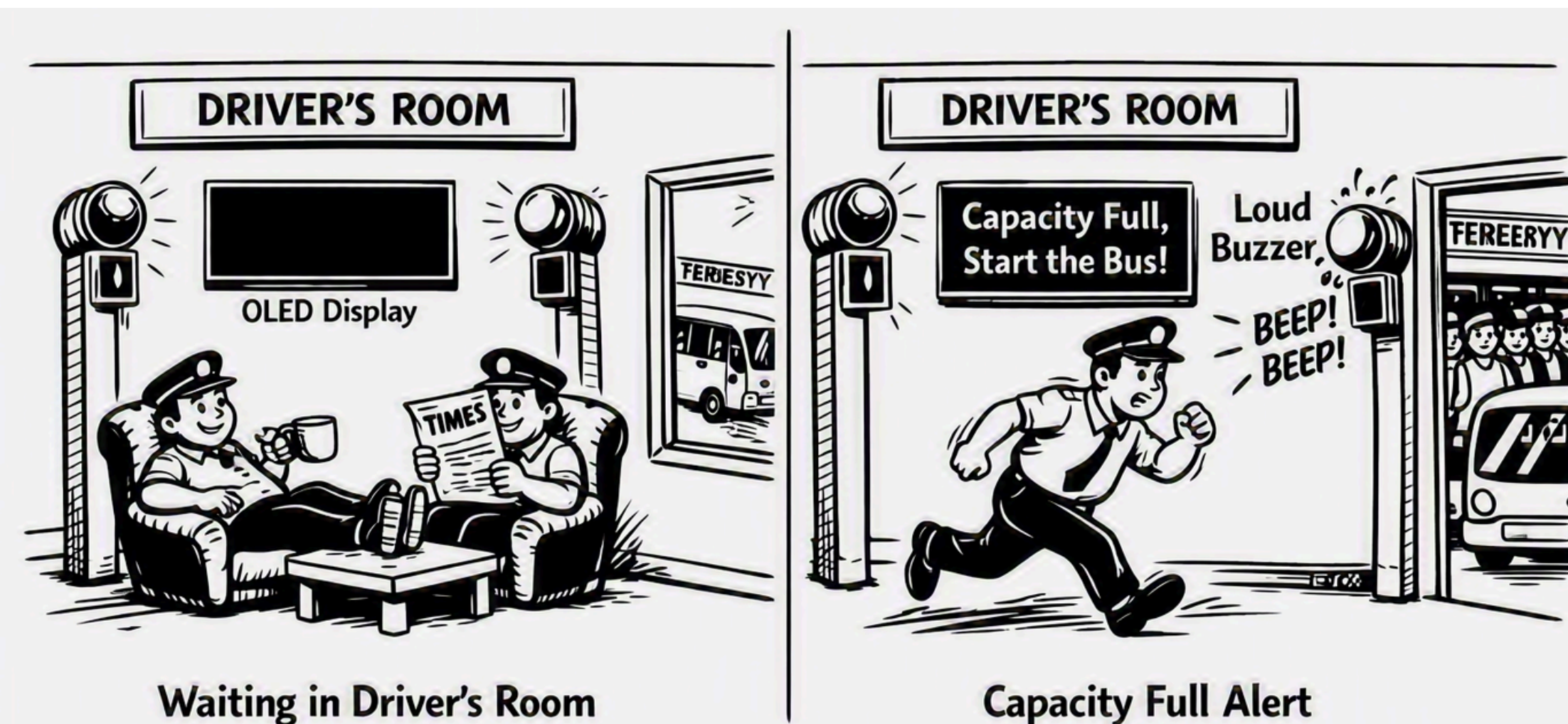
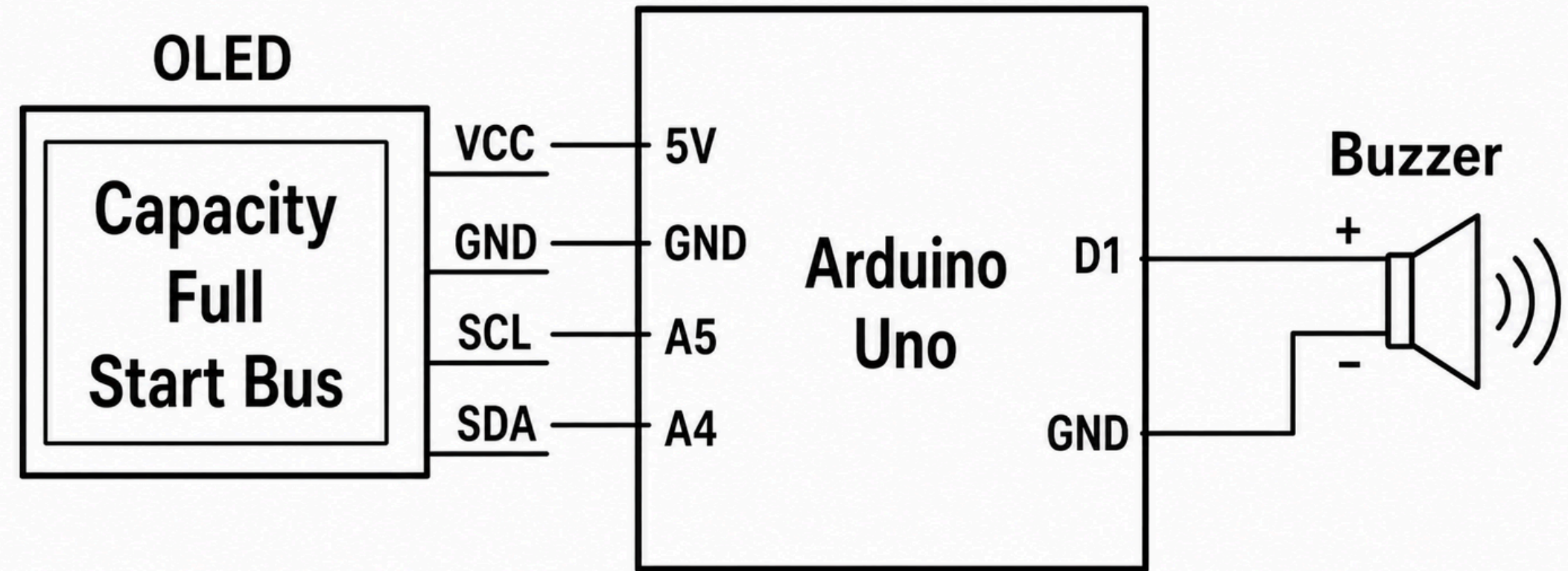
Empty Shelter



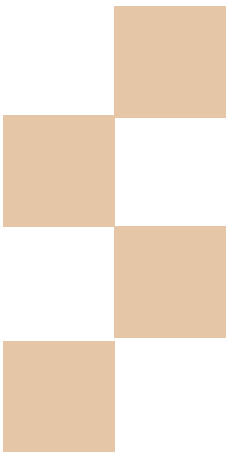
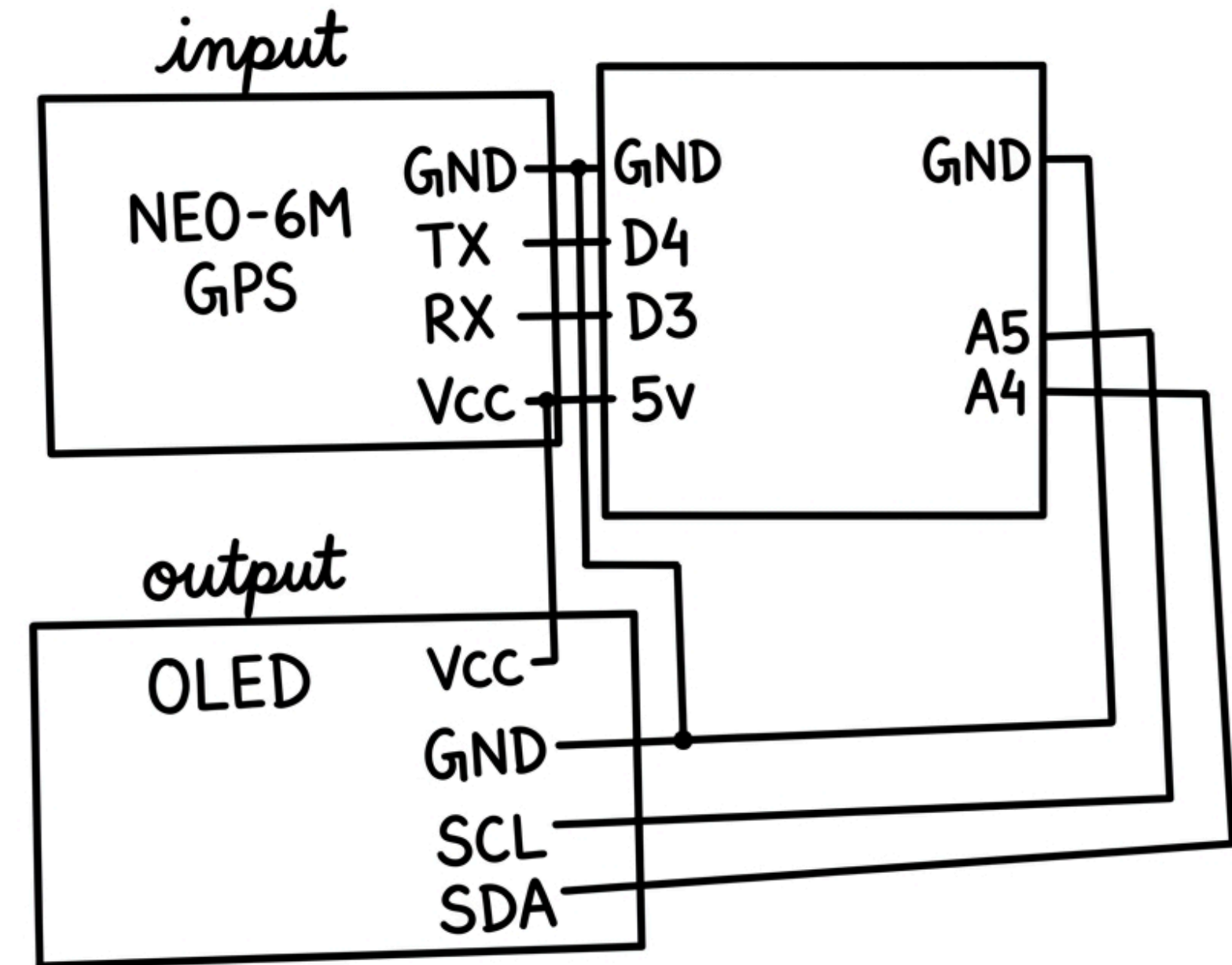
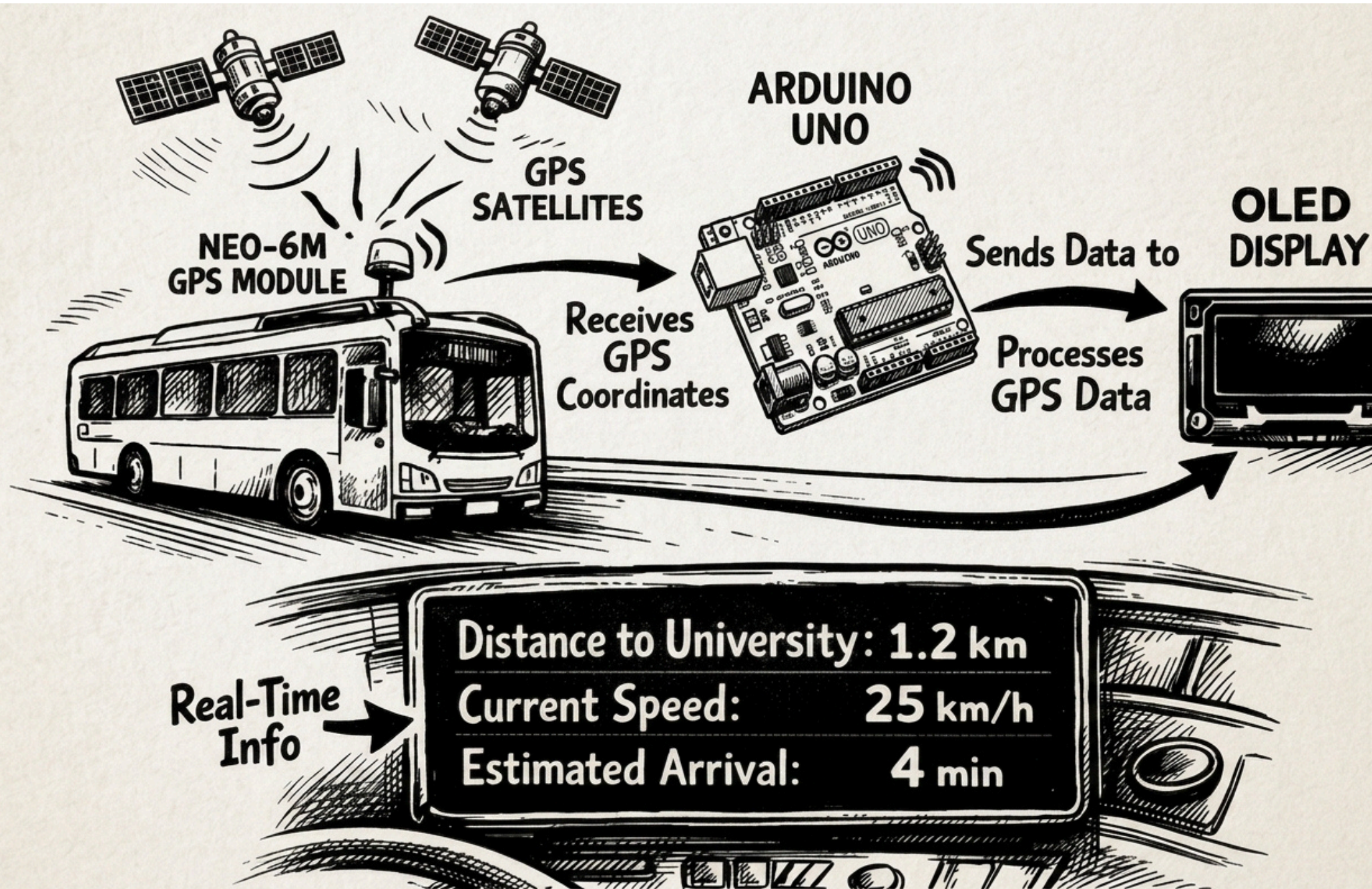
Full Shelter



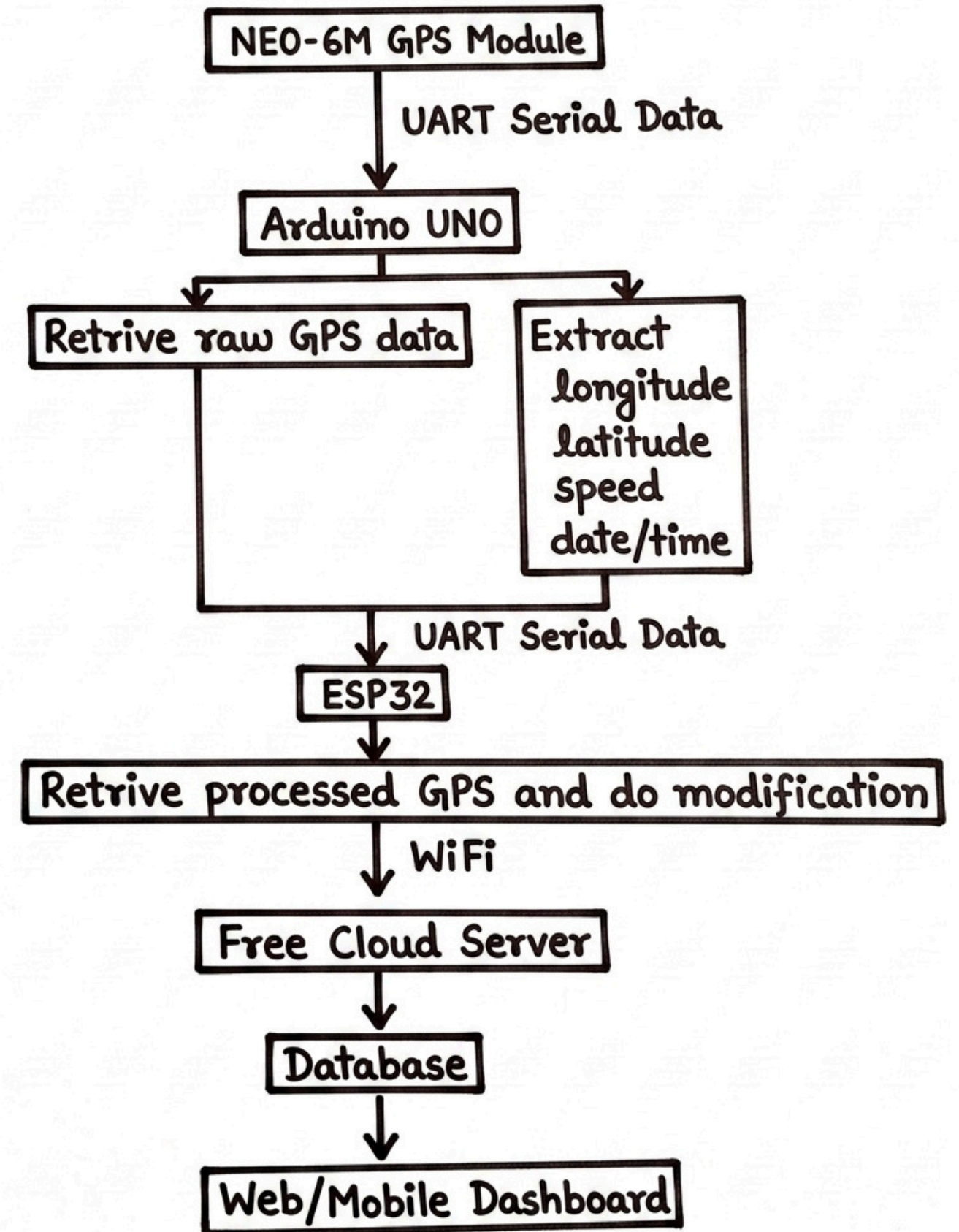
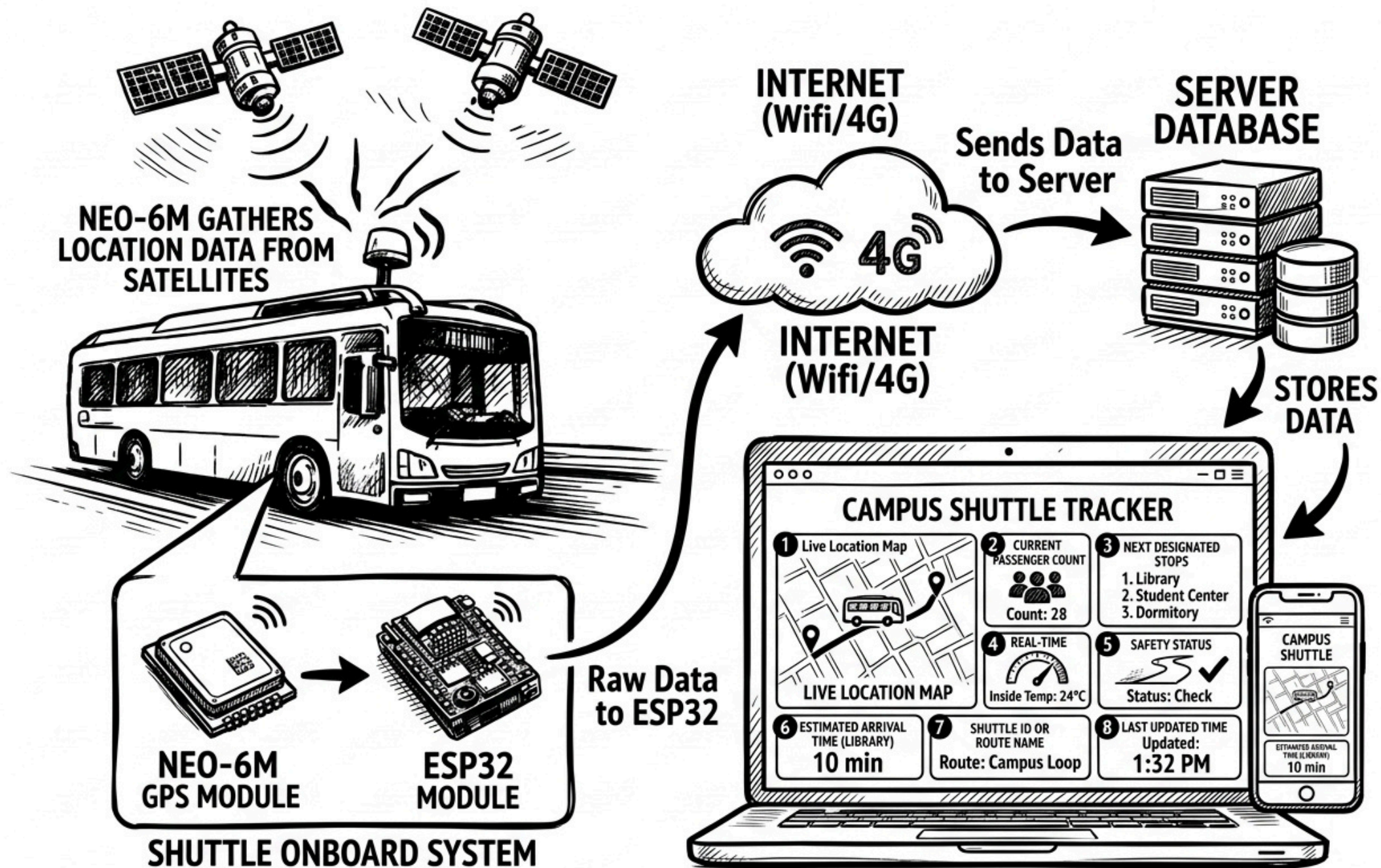
Feature 2



Feature 3



Feature 4



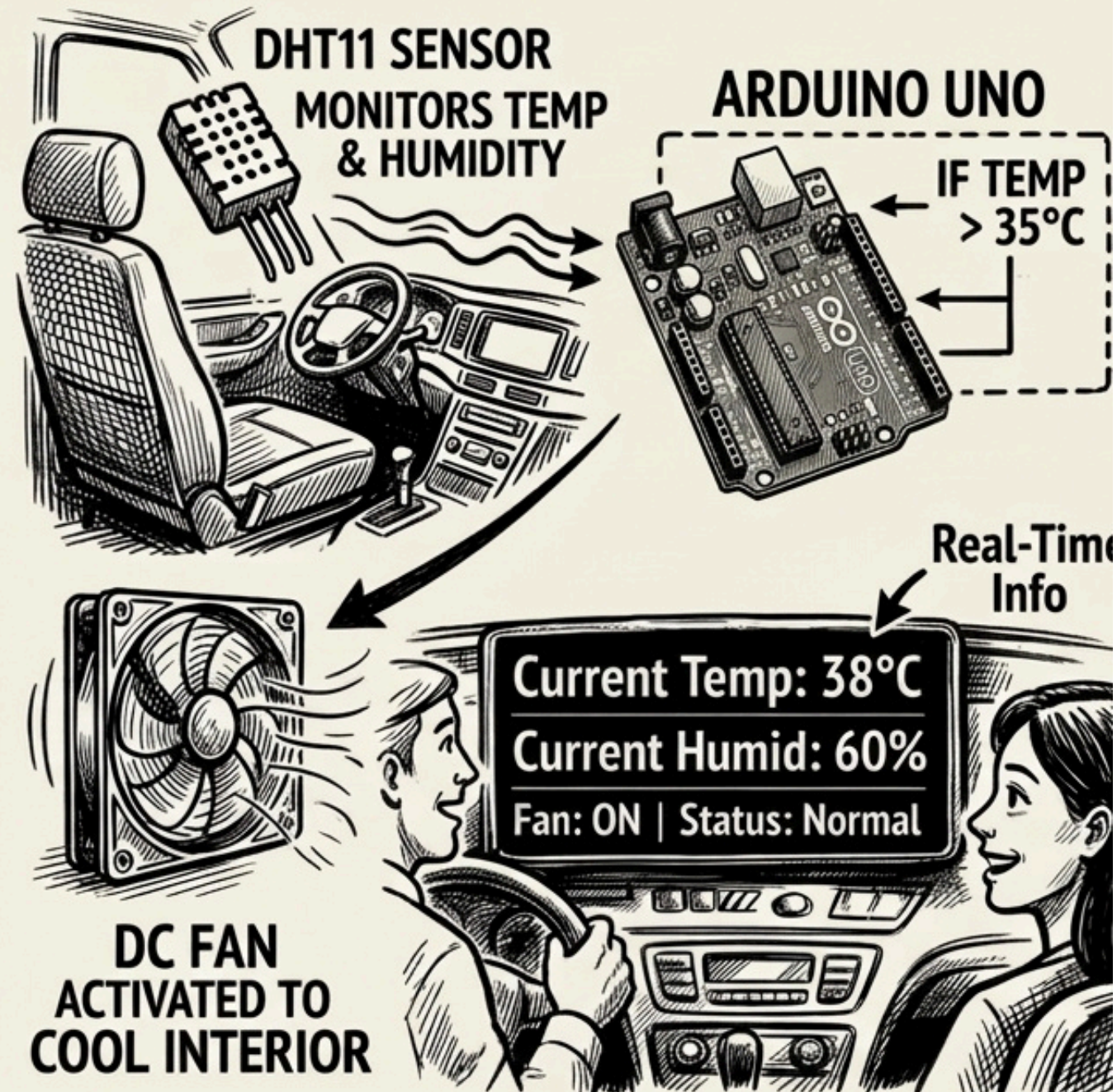
Feature 5

BUS INTERIOR SAFETY AND CLIMATE SYSTEM

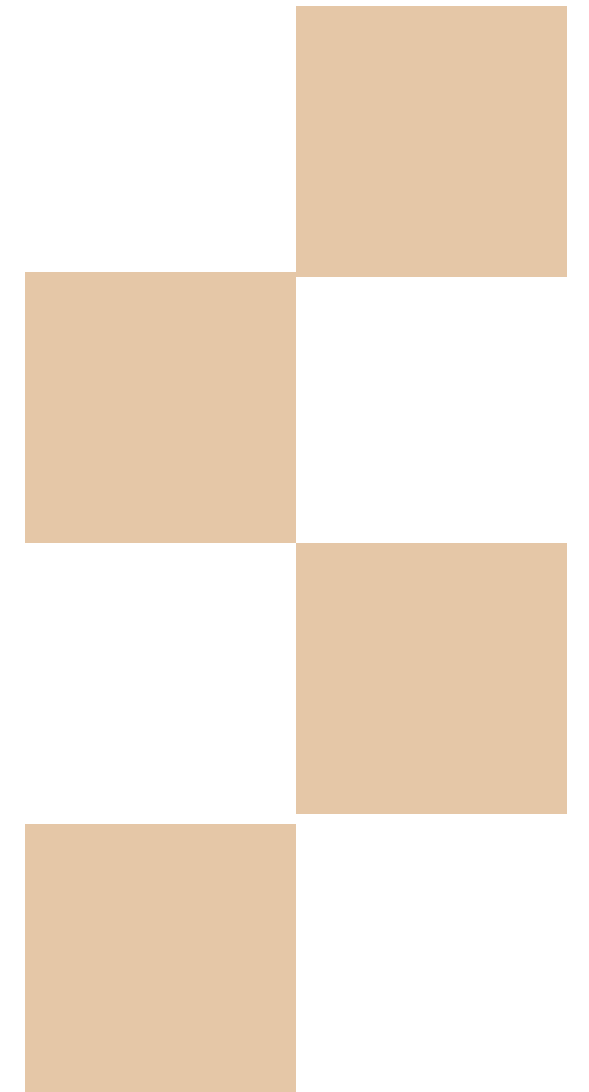
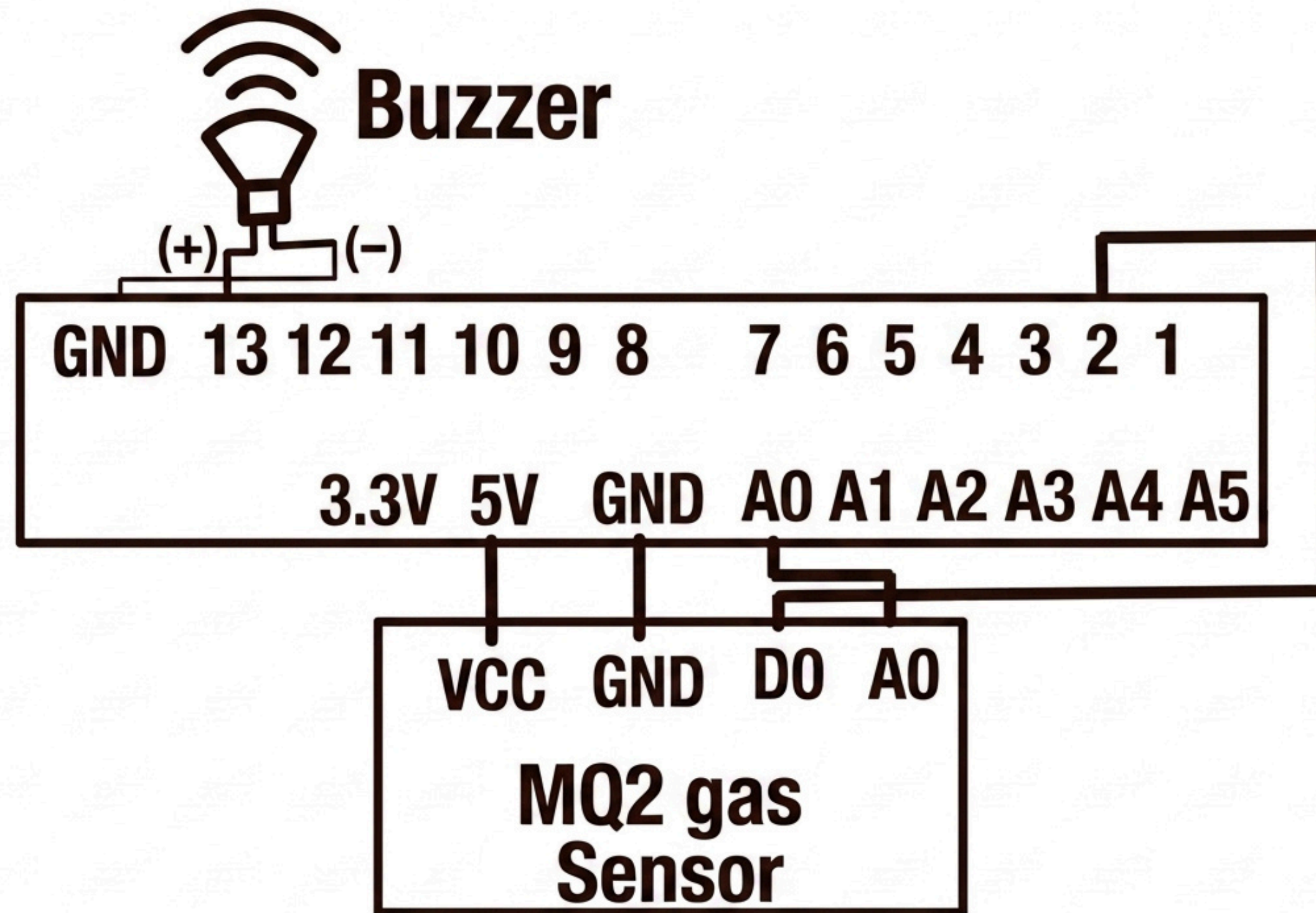
AIR QUALITY MONITORING & ALERT



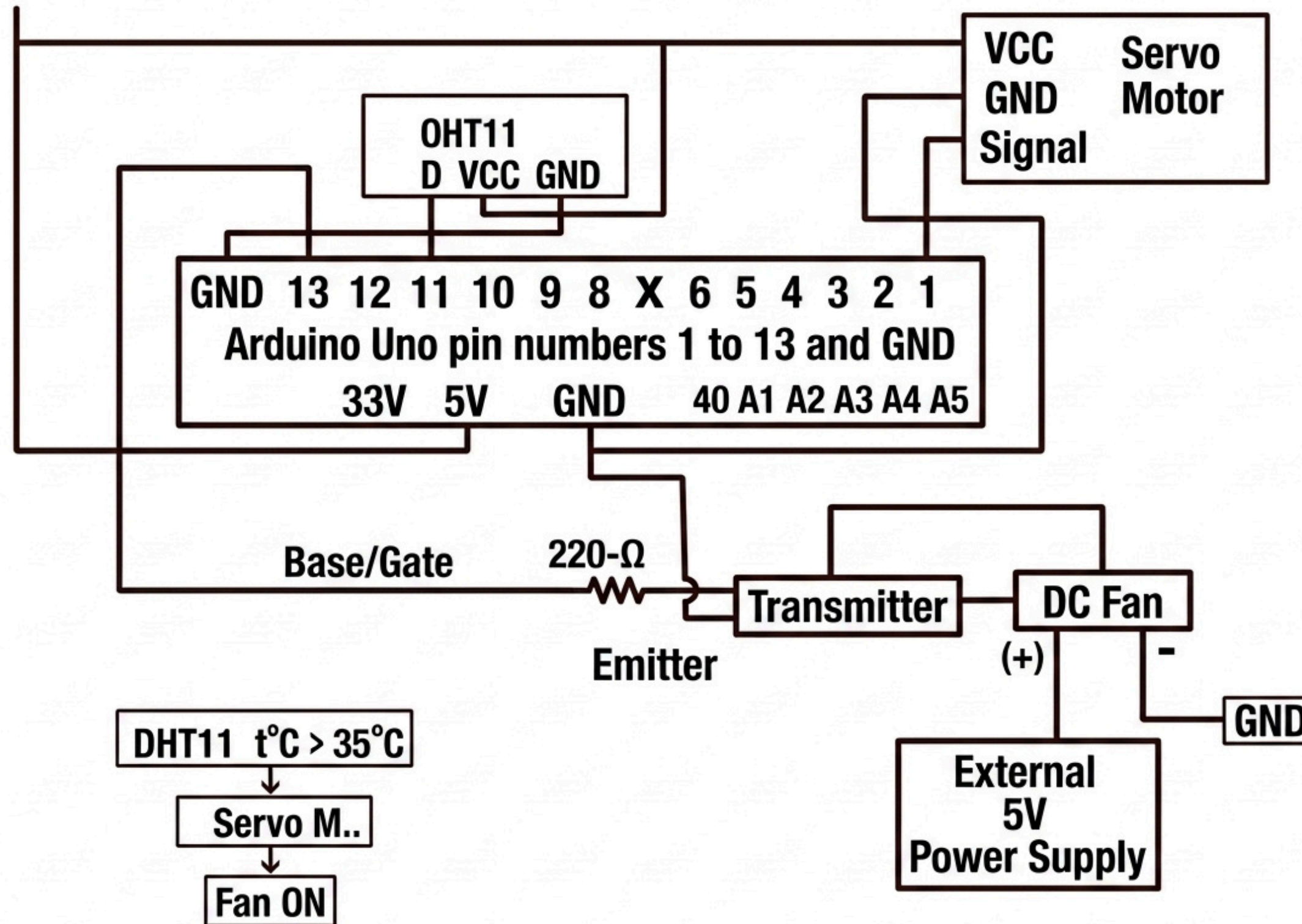
OLD & FAN: TEMPERATURE & HUMIDITY CONTROL



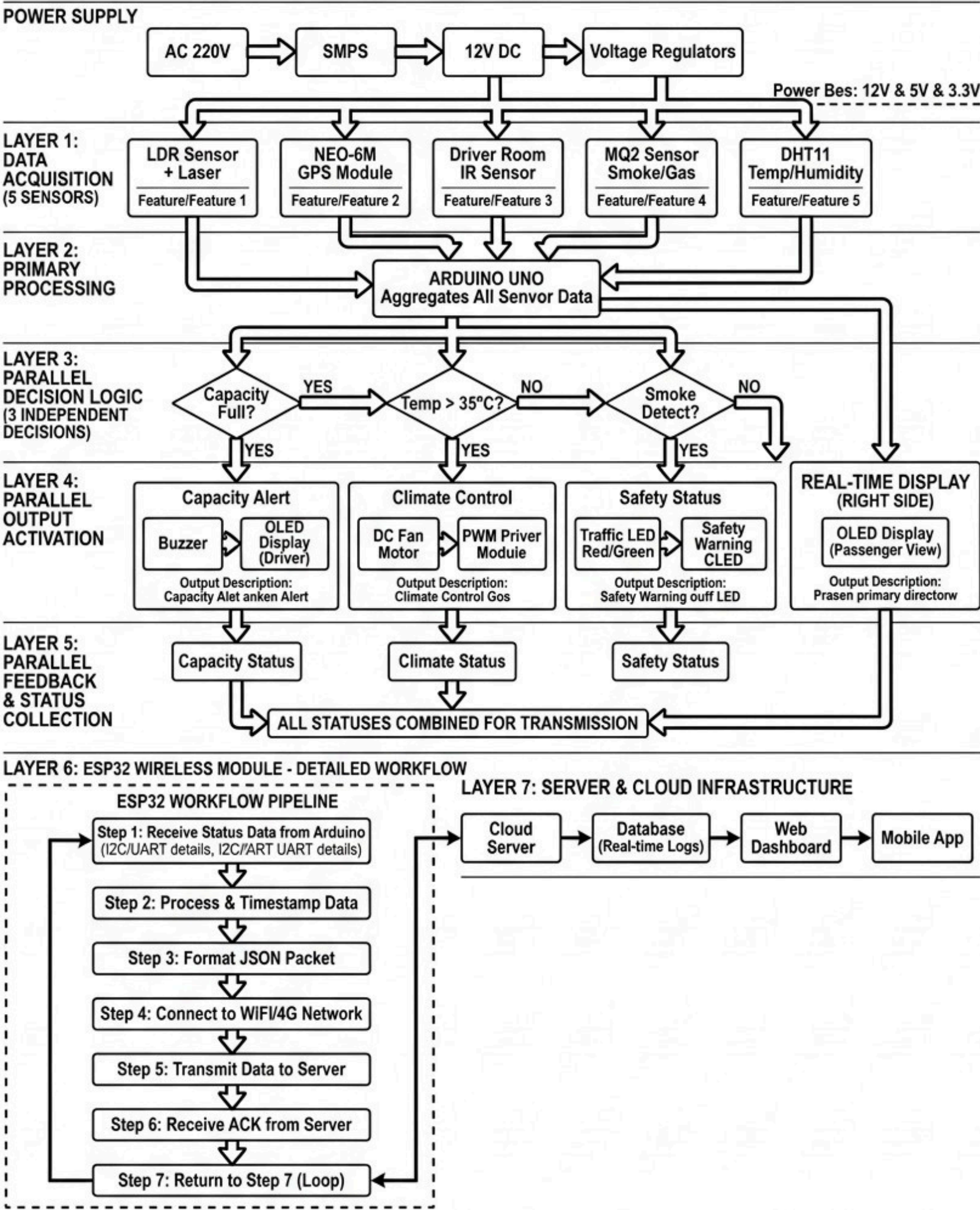
Feature 5



Feature 5



Diagram



- **Power Supply:** The system converts AC 220V power into regulated 12V, 5V, and 3.3V DC supplies to power the Arduino, ESP32, sensors, and output devices.

- **Data Acquisition:** Five components—LDR sensor with laser, NEO-6M GPS module, IR sensor, MQ2 smoke/gas sensor, and DHT11 temperature and humidity sensor—collect real-time shuttle data.

- **Primary Processing:** The Arduino UNO receives and processes data from all sensors and GPS modules, acting as the main controller of the system.

- **Parallel Decision and Output Control:** The Arduino makes independent decisions regarding shuttle capacity, temperature control, and smoke detection, activating the buzzer, OLED display, DC fan, traffic LED, and safety warning system when required.

- **Status and Real-Time Display:** All capacity, climate, and safety information is combined and transmitted to provide real-time passenger information through the OLED display.

- **ESP32 and Cloud Communication:** The ESP32 receives the processed status and GPS data from the Arduino, adds timestamps, formats the data into JSON, connects to Wi-Fi/4G, and sends it to the cloud server, database, web dashboard, and mobile application for remote monitoring.

Components



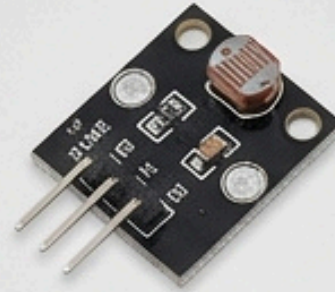
Arduino UNO R3



ESP32 Dev Board



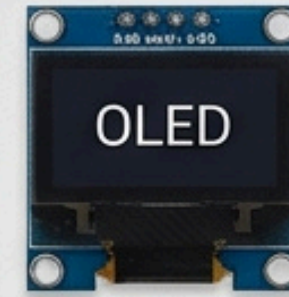
Laser Module



LDR Module



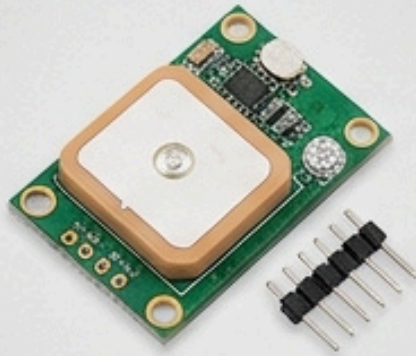
Red LED



0.96 inch OLED



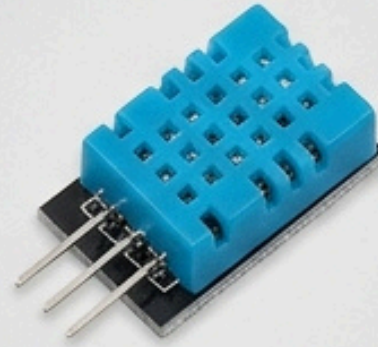
Piezo Buzzer



NEO-6M GPS



MQ2 Sensor



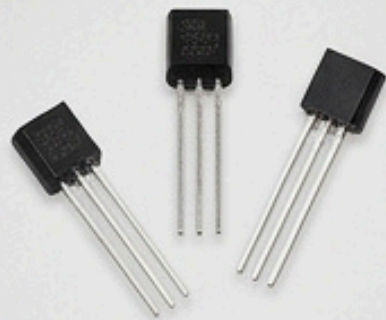
DHT11 Sensor



SG90 Servo Motor



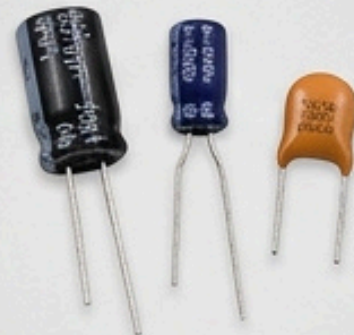
DC Fan



NPN Transistor
(2N2222)



Resistors



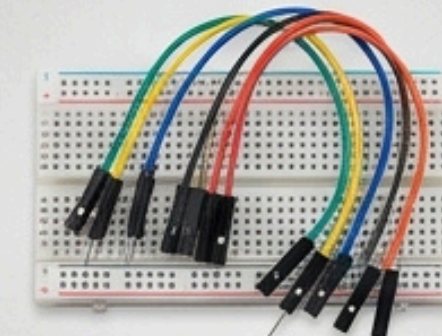
Capacitors



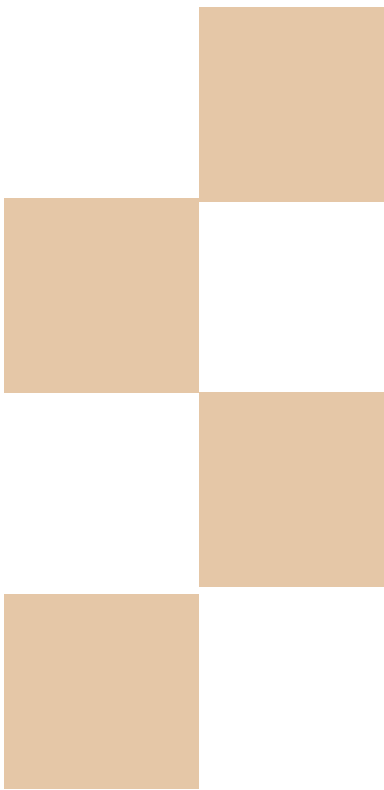
Diodes



USB to TTL
Converter



Jumper Wires
& Breadboard



Thank You